

Modelling Optical Degradation of the Liverpool Telescope

Anton Robb

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1 Overview

The Liverpool Telescope, operated by Liverpool John Moores University, is a 2-metre robotic telescope on La Palma in the Canary Islands. Every clear night, the total reflectivity of its optical system (the fraction of incident light that reaches the focus, referred to here as *transmission*) is measured. As atmospheric particles accumulate on the mirrors, transmission steadily declines; periodic cleaning and resurfacing restore it in sudden jumps.

This project models that degradation over roughly a decade of measurements. The aims are to identify when the mirrors were cleaned, fit a physically motivated model of the decay between cleanings, separate genuine measurements from cloud-affected ones, test whether the rate of degradation itself changes over time, and forecast when transmission will next fall below the operational threshold.

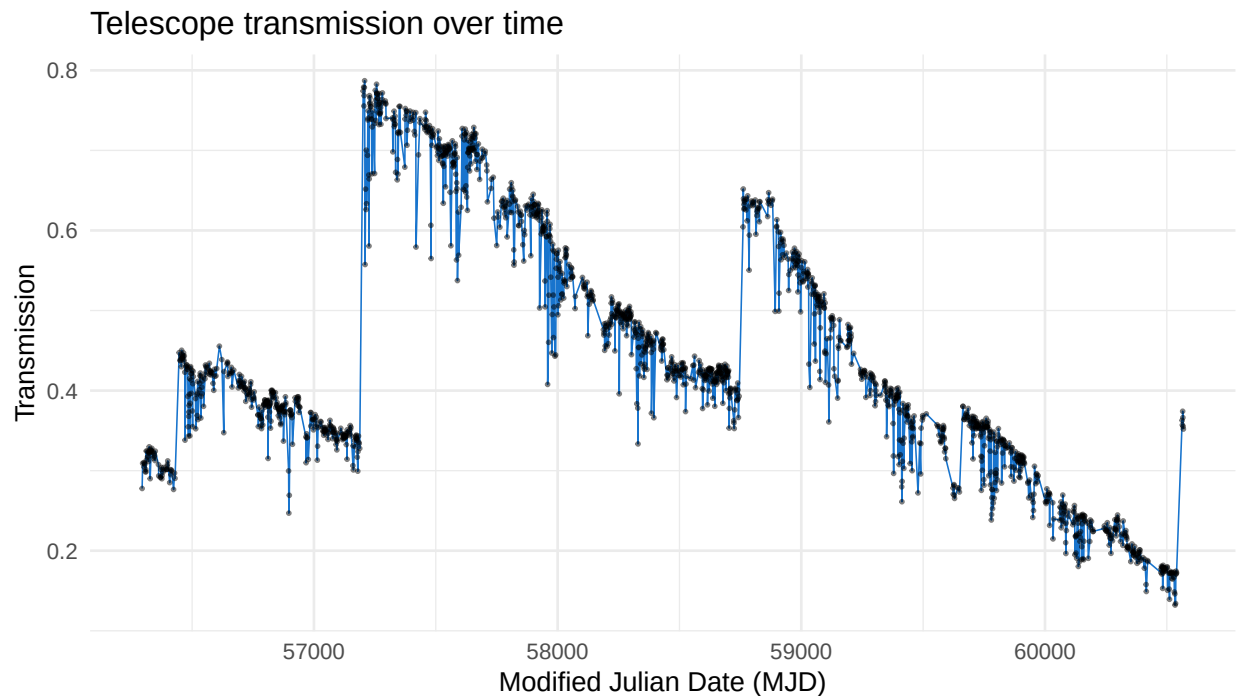
The dataset (`telescope.csv`) contains a date (as a Modified Julian Date, MJD) and a transmission value between 0 and 1 for each night, with NA on nights when no measurement was possible.

2 Data and initial inspection

```
library(ggplot2)
library(dplyr)

telescope <- read.csv("telescope.csv")
tel <- telescope[!is.na(telescope$trans), ]

ggplot(tel, aes(mjd, trans)) +
  geom_line(colour = "dodgerblue3", linewidth = 0.3) +
  geom_point(size = 0.4, alpha = 0.5) +
  labs(title = "Telescope transmission over time",
       x = "Modified Julian Date (MJD)", y = "Transmission") +
  theme_minimal()
```



Two features are immediately visible. First, transmission trends downward within each interval and then jumps sharply upward at cleaning or resurfacing events. Second, a scattering of anomalously low points sits well below the main trend: these correspond to partly cloudy nights, where cloud cover reduces the *measured* transmission without the telescope's true reflectivity having changed. Both features shape the modelling that follows.

3 Identifying cleaning events

Cleanings appear as abrupt upward steps in transmission. Inspecting the series, the clearest steps occur at approximately the following MJDs, which I take as the cleaning boundaries. (A cleaning near the very end of the record is also apparent.)

```
cleans <- c(56430, 57188, 58745, 60540)
```

These boundaries divide the record into segments, within each of which transmission is expected to decay smoothly.

4 A physical model of decay between cleanings

Between cleanings, contamination accumulates and transmission is expected to decline roughly exponentially in time. Taking the logarithm of transmission linearises this exponential decay, so I model $\log(\text{transmission})$ as a linear function of time within each segment, allowing each segment its own intercept (the recovery at each cleaning) while sharing a common slope.

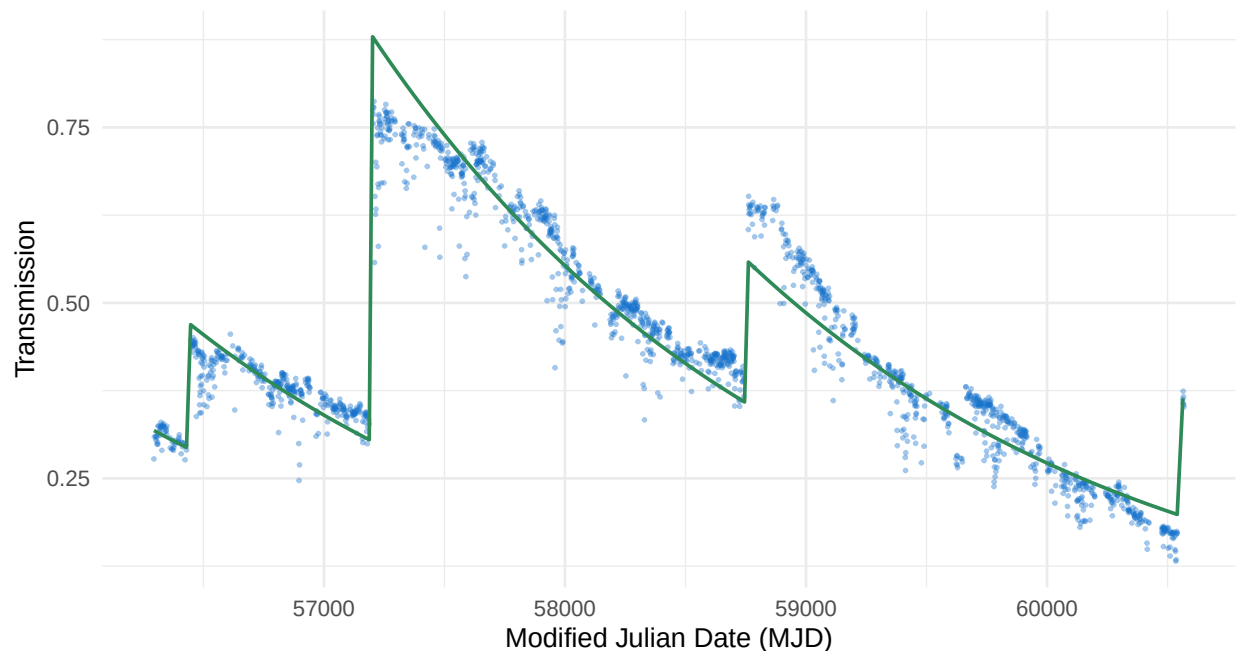
```
tel$log_trans <- log(tel$trans)
tel$segment <- cut(tel$mjd,
                  breaks = c(min(tel$mjd), cleans, max(tel$mjd)),
                  include.lowest = TRUE)

model <- lm(log_trans ~ mjd + segment, data = tel)

tel$fit <- exp(predict(model))

ggplot(tel, aes(mjd, trans)) +
  geom_point(size = 0.4, alpha = 0.4, colour = "dodgerblue3") +
  geom_line(aes(y = fit), colour = "seagreen", linewidth = 0.7) +
  labs(title = "Transmission with fitted exponential-decay model",
       x = "Modified Julian Date (MJD)", y = "Transmission") +
  theme_minimal()
```

Transmission with fitted exponential-decay model

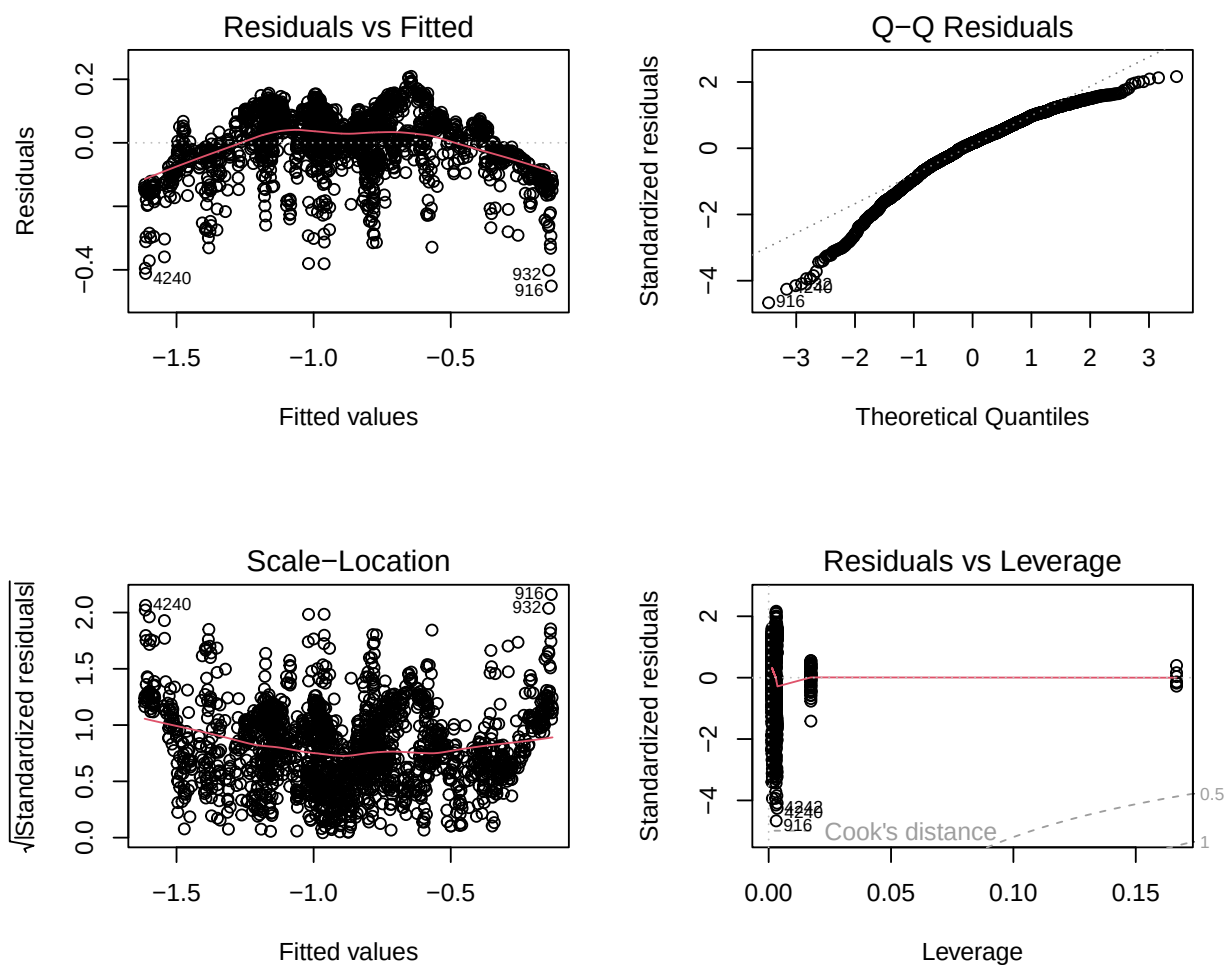


The model captures the broad decay-and-recover pattern, but it is visibly pulled downward by the cloud-affected points, which the next section addresses.

5 Separating cloud-affected measurements

Cloud cover can only *reduce* measured transmission, never inflate it, so cloud-affected points appear as large negative departures from the model. A model check plot makes this concrete.

```
par(mfrow = c(2, 2))
plot(model)
```



```
par(mfrow = c(1, 1))
```

The effect is clearest in the Residuals vs Fitted and Q-Q panels, where a heavy one-sided negative tail appears: many points fall far below the fit, with no matching positive tail, exactly the signature of a process that only ever depresses the measurement. The Residuals vs Leverage panel shows these points have low leverage, confirming they are measurement artefacts rather than influential observations reflecting true changes in mirror performance.

5.1 Iterative outlier removal

A single one-pass cut is sensitive to the fact that the cloud-affected points themselves distort the model they are being compared against: while they remain in the fit, they inflate the residual spread and pull the fitted line down, so a threshold defined in standard deviations is too lenient and leaves cloud points behind. I therefore remove them iteratively. At each step I refit the model on the currently retained data, recompute residuals, and flag points falling more than a fixed number of standard deviations *below* the fit (a one-sided cut, since only downward departures indicate cloud). Repeating until the retained set stabilises lets the model tighten around the genuine data and progressively expose cloud points that a single pass would miss.

```
# One-sided threshold in residual standard deviations (below the fit only)
k <- 2

keep <- rep(TRUE, nrow(tel))
repeat {
  m <- lm(log_trans ~ mjd + segment, data = tel[keep, ])
  resid_all <- tel$log_trans - predict(m, newdata = tel)
  s <- sd(resid_all[keep])
  new_keep <- resid_all > -k * s # cloud points sit far BELOW the fit
  if (identical(new_keep, keep)) break
  keep <- new_keep
}

tel$keep <- keep
model_clean <- lm(log_trans ~ mjd + segment, data = tel[tel$keep, ])

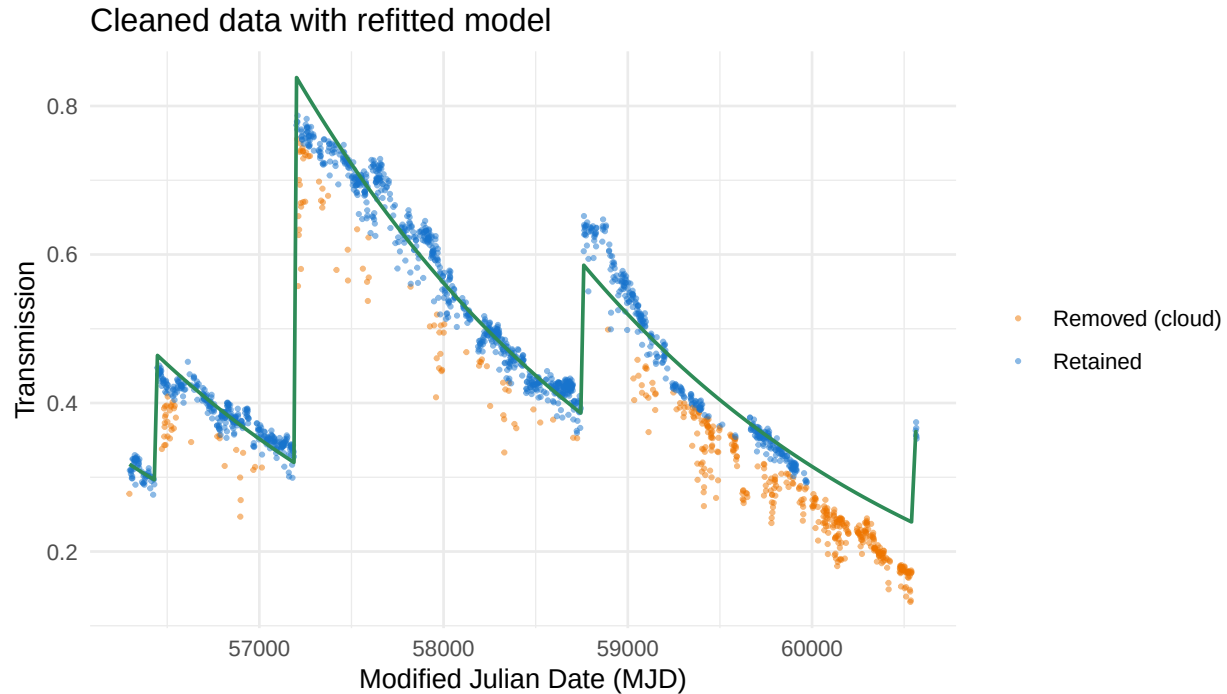
n_removed <- sum(!tel$keep)
n_retained <- sum(tel$keep)
c(removed = n_removed, retained = n_retained)
```

```
## removed retained
##      490      1422
```

Iterating to convergence removes more cloud-affected points than a single pass at the same threshold, because each refit tightens the residual scale and reveals further outliers that the initial, inflated spread had masked.

```
tel$pred_trans <- exp(predict(model_clean, newdata = tel))

ggplot(tel, aes(mjd, trans, colour = keep)) +
  geom_point(size = 0.5, alpha = 0.5) +
  geom_line(aes(y = pred_trans), colour = "seagreen", linewidth = 0.7) +
  scale_colour_manual(values = c("TRUE" = "dodgerblue3", "FALSE" = "darkorange2"),
    labels = c("TRUE" = "Retained", "FALSE" = "Removed (cloud)"),
    name = NULL) +
  labs(title = "Cleaned data with refitted model",
    x = "Modified Julian Date (MJD)", y = "Transmission") +
  theme_minimal()
```



6 Does the degradation rate change over time?

A natural question is whether the mirrors degrade at a constant rate or whether the rate itself changes. Testing this correctly requires care, because transmission measurements are a time series: readings close together in time are correlated (short-term trends over timescales of up to about a month), so the nightly observations are not independent. Treating them as independent, as a naive regression would, badly understates the standard errors and produces over-confident p-values, because the *effective* number of independent observations is far smaller than the raw count of nights.

6.1 Accounting for non-independence

To address this, I aggregate the cleaned nightly data to monthly means before testing for a change in rate. Averaging within calendar months collapses the short-term autocorrelation into a series whose points are much closer to independent, so the resulting inference reflects the genuine effective sample size rather than the inflated nightly count. This is a deliberately conservative approach: it discards some within-month information in exchange for honest standard errors and p-values.

```
library(lubridate)

# Reconstruct a calendar date from MJD (MJD epoch: 1858-11-17)
tel_clean <- tel[tel$keep, ]
tel_clean$date <- as.Date(tel_clean$mjd, origin = as.Date("1858-11-17"))
tel_clean$month <- floor_date(tel_clean$date, "month")

monthly <- tel_clean %>%
  group_by(month, segment) %>%
  summarise(mjd = mean(mjd), log_trans = mean(log_trans), n = n(),
    .groups = "drop")
```

With a near-independent monthly series, I compare a constant-rate model (linear in time) against one allowing the rate to change (adding a quadratic term in time).

```
m_linear <- lm(log_trans ~ mjd + segment, data = monthly)
m_quad   <- lm(log_trans ~ mjd + I(mjd^2) + segment, data = monthly)

anova(m_linear, m_quad)
```

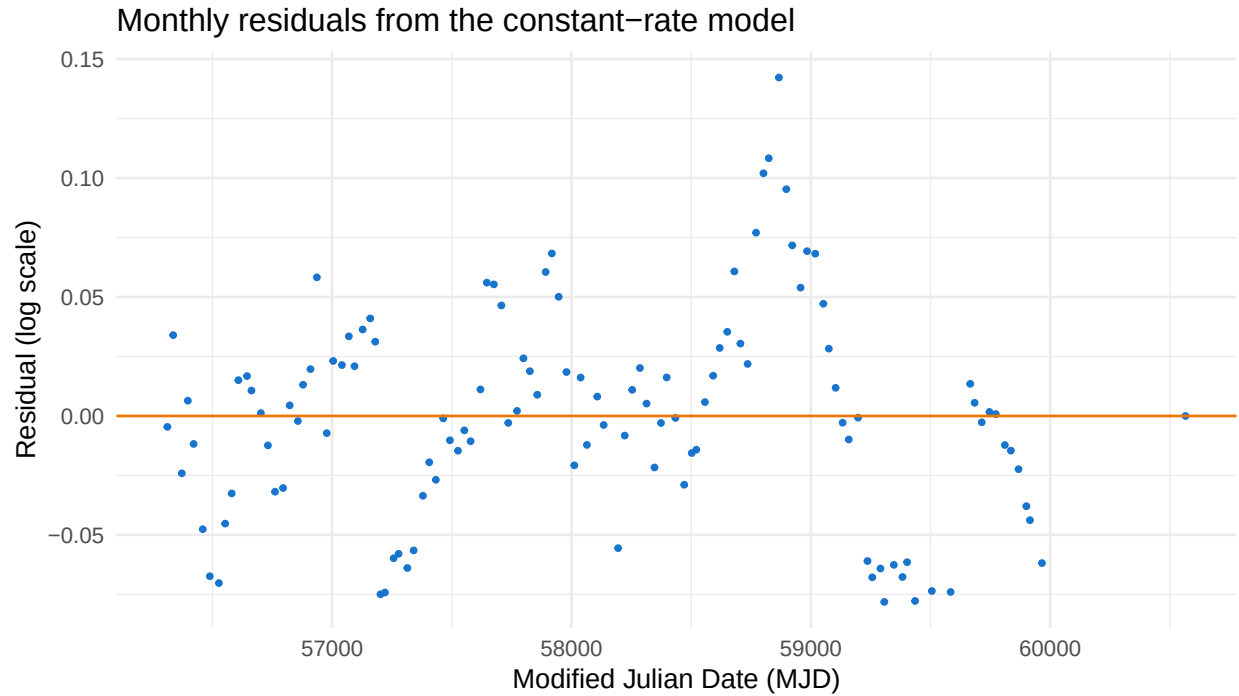
```
## Analysis of Variance Table
##
## Model 1: log_trans ~ mjd + segment
## Model 2: log_trans ~ mjd + I(mjd^2) + segment
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     112 0.22705
## 2     111 0.14847   1  0.078582 58.75 7.29e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_quad)$coefficients["I(mjd^2)", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## -4.463738e-08  5.823657e-09 -7.664837e+00  7.290183e-12
```

Fitting on the monthly-aggregated series, the quadratic term tests whether the degradation rate is genuinely changing over time rather than reflecting autocorrelated nightly noise. Because the standard errors here are based on a realistic effective sample size, any significance in the quadratic term is a credible indication of a secular change in the degradation rate, not an artefact of treating correlated nightly readings as independent.

```
ggplot(data.frame(mjd = monthly$mjd, r = residuals(m_linear)),
  aes(mjd, r)) +
  geom_point(colour = "dodgerblue3", size = 0.8) +
  geom_hline(yintercept = 0, colour = "darkorange2") +
  labs(title = "Monthly residuals from the constant-rate model",
    x = "Modified Julian Date (MJD)", y = "Residual (log scale)") +
  theme_minimal()
```



Structure in these residuals over long timescales (as opposed to random scatter) would corroborate a changing degradation rate; the quadratic test quantifies it.

7 Forecasting

7.1 Transmission on a target date

Using the model fitted to the cleaned, aggregated data, I predict transmission on 1 January 2026 (MJD 61041) with a 95% confidence interval, back-transforming from the log scale.

```
latest_seg <- tail(levels(monthly$segment), 1)
newdata <- data.frame(mjd = 61041,
                      segment = factor(latest_seg, levels = levels(monthly$segment)))

pred <- predict(m_quad, newdata = newdata, interval = "confidence", level = 0.95)
exp(pred)
```

```
##      fit      lwr      upr
## 1 0.255184 0.2360335 0.2758884
```

7.2 When will transmission next fall below 15%?

The telescope's operational requirement is that transmission stays above 15%. Assuming no further cleaning, I project the model forward to find the date at which predicted transmission crosses this threshold, with a confidence interval derived from the model's prediction bounds.


```

future_mjd <- seq(max(monthly$mjd), max(monthly$mjd) + 2000, by = 1)
future <- data.frame(mjd = future_mjd,
                     segment = factor(latest_seg, levels = levels(monthly$segment)))

fp <- exp(predict(m_quad, newdata = future, interval = "confidence", level = 0.95))
threshold <- 0.15

cross <- data.frame(
  fit = future_mjd[which(fp[, "fit"] < threshold)[1]],
  lower = future_mjd[which(fp[, "upr"] < threshold)[1]],
  upper = future_mjd[which(fp[, "lwr"] < threshold)[1]]
)
cross

##          fit      lower      upper
## 1 61720.5 61862.5 61596.5

as.Date(unlist(cross), origin = as.Date("1858-11-17"))

##          fit          lower          upper
## "2027-11-11" "2028-04-01" "2027-07-10"

```

The point estimate and interval give the expected date, and its uncertainty, at which the mirrors would need attention to stay within the operational requirement.

8 Summary

Transmission between cleanings follows a clear exponential decay, well captured by a log-linear model segmented at the cleaning events. Cloud-affected nights, which only ever depress the measurement, are best removed iteratively: refitting after each cut tightens the model and exposes further cloud points that a single pass leaves behind. Crucially, testing whether the degradation rate changes over time requires accounting for the strong autocorrelation of nightly measurements; aggregating to monthly means before inference yields an honest effective sample size and avoids the over-confident conclusions a naive nightly regression would produce. The final model supports forecasting both the transmission on a given date and the date at which cleaning will next be required.

9 Notes

Data: nightly transmission measurements from the Liverpool Telescope, provided by Liverpool John Moores University and included here with permission. This project began as a piece of university coursework and has been reworked as an independent analysis.